Dataset Summary

The dataset contains **8,173 Bengaluru traffic incident records** collected between **2023–2024** across **22 corridors** and **294 junctions**.

Incident Distribution

The data is heavily dominated by operational road incidents:

- Vehicle Breakdown: **4,896 (~60%)**
- Others: **638**
- Pot Holes: **537**
- Construction: **480**
- Water Logging: **458**
- Accident: **365**
- Tree Fall: **284**
- Road Conditions: **170**
- Congestion: **136**
- Public Events: **84**

A total of **17 unique incident causes** are present.

Geographic Coverage

- **22 traffic corridors**
- **294 junctions**
- Latitude and longitude available for spatial analysis and clustering.

Operational Metadata Available

Each incident includes:

- Event cause/type
- Corridor and junction
- Vehicle type (when applicable)
- Road closure requirement
- Priority level
- Start, resolution, and closure timestamps
- Zone and police station information

Target Variable Construction

Incident duration is derived as:

`duration_mins = closed_datetime - start_datetime`

Out of 8,173 incidents, only **3,129 records contain valid closure timestamps**, making them usable for supervised duration prediction.

Data Quality Observations

The duration labels contain significant noise:

- Median duration: **64 minutes**
- 75th percentile: **398 minutes**
- Maximum duration: **201,789 minutes (~140 days)**
- Negative duration records detected
- **666 incidents exceed 24 hours**

Therefore, duration labels require cleaning before model training. Records with negative durations should be removed, and extreme outliers should be capped or filtered (e.g., >1440 minutes).

Why This Dataset Fits Tree-Based ML Models

The dataset is primarily **structured tabular data**, not sequential time-series data. Prediction depends on interactions between:

- Event cause
- Corridor
- Hour of day
- Day of week
- Vehicle type
- Priority
- Road closure status

Because of these mixed categorical and temporal features, **XGBoost or Random Forest** are appropriate choices for predicting incident duration and disruption severity.